

Customer Experiences: Digital Transformation, Downtime-Optimized RISE Migrations & AI in Action

Nikhil Shetty, SAP

September 2026

Disclaimer

The information in this presentation is confidential and proprietary to SAP and may not be disclosed without the permission of SAP. Except for your obligation to protect confidential information, this presentation is not subject to your license agreement or any other service or subscription agreement with SAP. SAP has no obligation to pursue any course of business outlined in this presentation or any related document, or to develop or release any functionality mentioned therein.

This presentation, or any related document and SAP's strategy and possible future developments, products and or platforms directions and functionality are all subject to change and may be changed by SAP at any time for any reason without notice. The information in this presentation is not a commitment, promise or legal obligation to deliver any material, code or functionality. This presentation is provided without a warranty of any kind, either express or implied, including but not limited to, the implied warranties of merchantability, fitness for a particular purpose, or non-infringement. This presentation is for informational purposes and may not be incorporated into a contract. SAP assumes no responsibility for errors or omissions in this presentation, except if such damages were caused by SAP's intentional or gross negligence.

All forward-looking statements are subject to various risks and uncertainties that could cause actual results to differ materially from expectations. Readers are cautioned not to place undue reliance on these forward-looking statements, which speak only as of their dates, and they should not be relied upon in making purchasing decisions.

01 • Digital Transformations

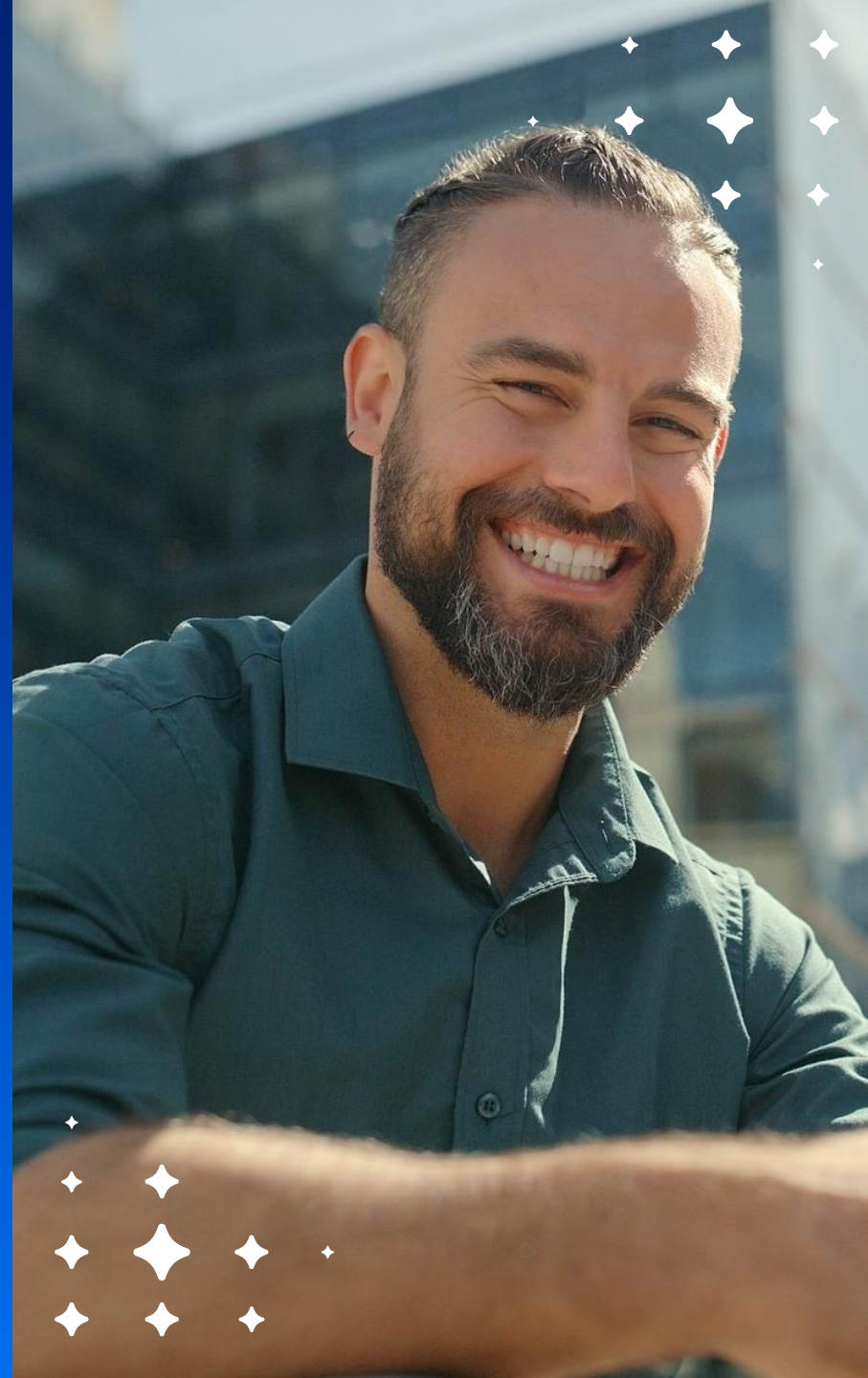
02 • Customer Exp – doDMO

03 • Customer Exp – ZDO

04 • Customer Exp – doC (Cust on TEMP)

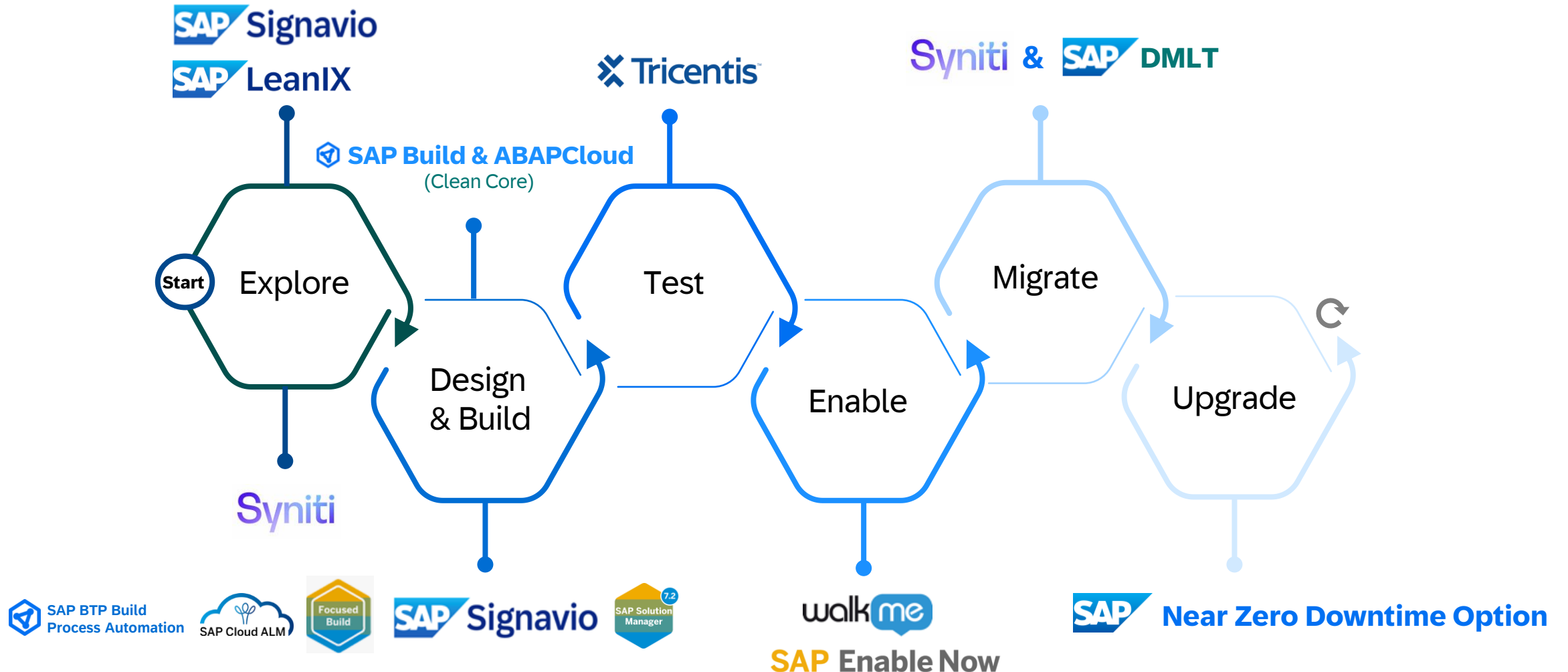
05 • AI in Action

Digital Transformation



The Big Picture: SAP Tools supporting an accelerated S/4HANA Roll-out

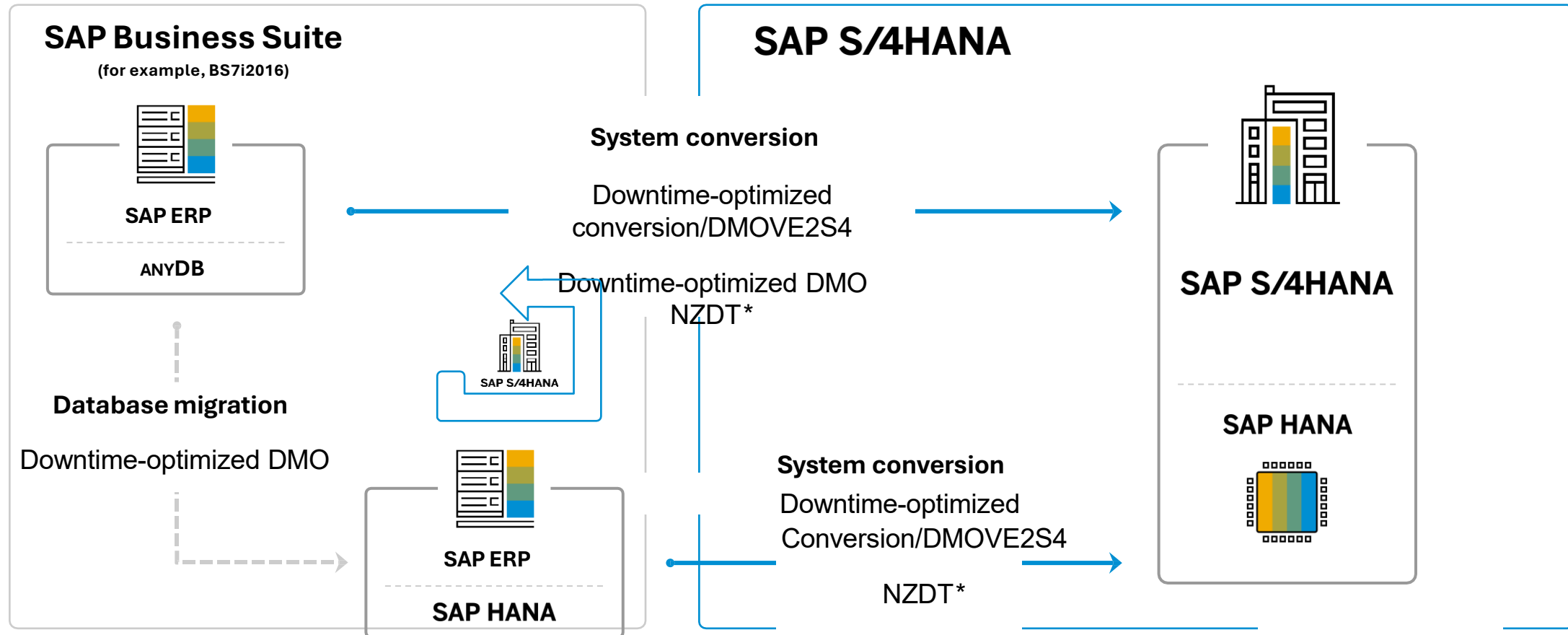
Systematically bridging Business and IT Transformation



Customer Experience – Downtime Optimized Maintenance



Downtime optimization approaches for system maintenance



* NZDT is service-based and currently being redesigned

Update, upgrade
nZDM and ZDO

Downtime-optimization approaches for maintenance activities

Approach	Abbreviation	Scenario	Availability*	SAP Note
near-Zero Downtime Maintenance (ABAP) **	nZDM (ABAP)	Update/Upgrade	Unrestricted available	1678565
Zero Downtime Option	ZDO	Update/Upgrade	Pilot Unrestricted available	Business Suite: 2163060 SAP S/4HANA: 2707731
downtime-optimized Database Migration Option	downtime-optimized DMO	Migration to SAP HANA database	Unrestricted available	2547309
downtime-optimized Conversion	doC	Conversion to SAP S/4HANA	Available for educated experts	doC Portal Page
Near Zero Downtime Technology	NZDT	Multiple (such as SAP S/4HANA Conversion)	Service-based	693168

* This is the current state of planning and may be changed by SAP at any time.

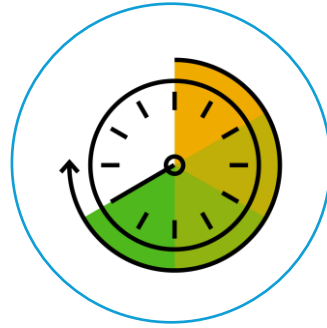
** This is different to nZDM for SAP HANA database

SAP S/4HANA conversion methodologies



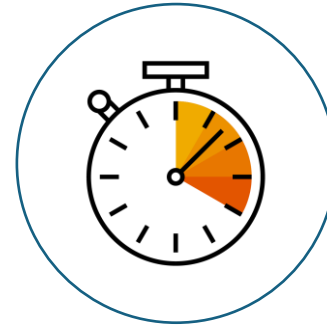
- **Standard approach**

- General availability
- SAP Note [3106931](#)



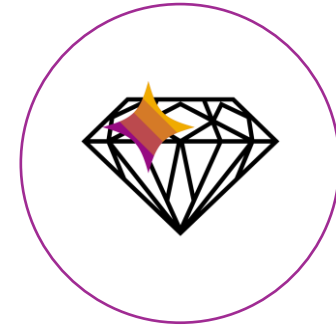
- **downtime-optimized Database Migration Option**

- General availability
- SAP Note [2547309](#)



- **downtime-optimized Conversion**

- General availability
- [doC Portal Page](#)



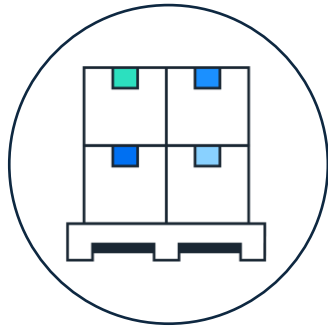
- **Near-Zero Downtime Technology**

- Minimized Downtime Service
- SAP Note [693168](#)

EFFORT

DOWNTIME

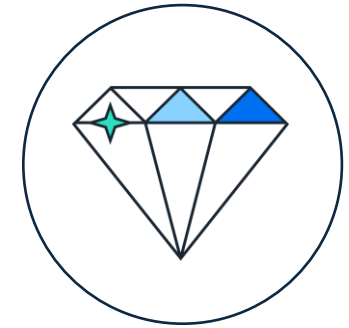
Options for updating SAP S/4HANA using the Software Update Manager (SUM)



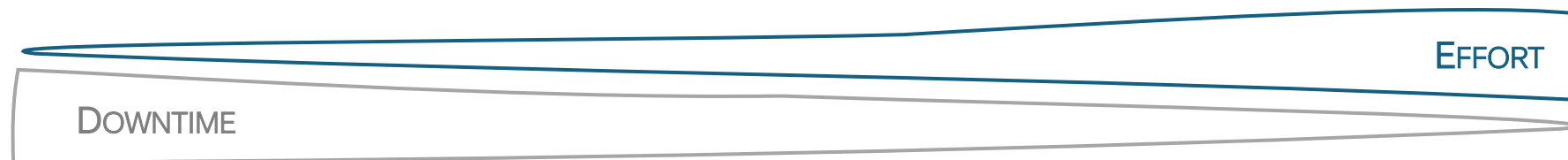
- Standard approach
- Several downtime optimizations (for example, shadow system operations)



- Near-Zero Downtime Maintenance (nZDM)
- Downtime reduction by moving downtime-relevant phases to uptime

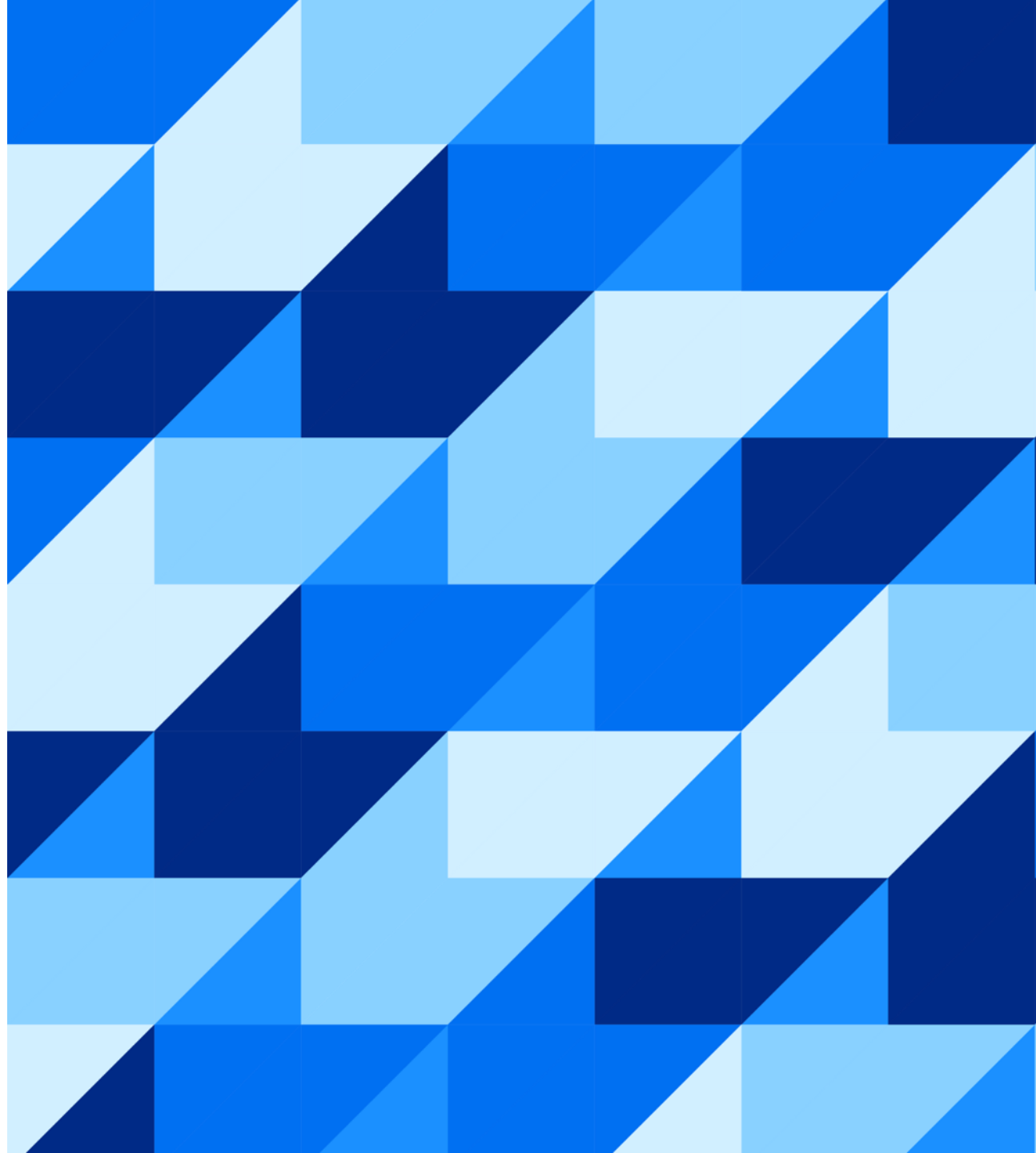


- Zero Downtime Option (ZDO)
- Complete upgrade running in parallel with continuous business operations



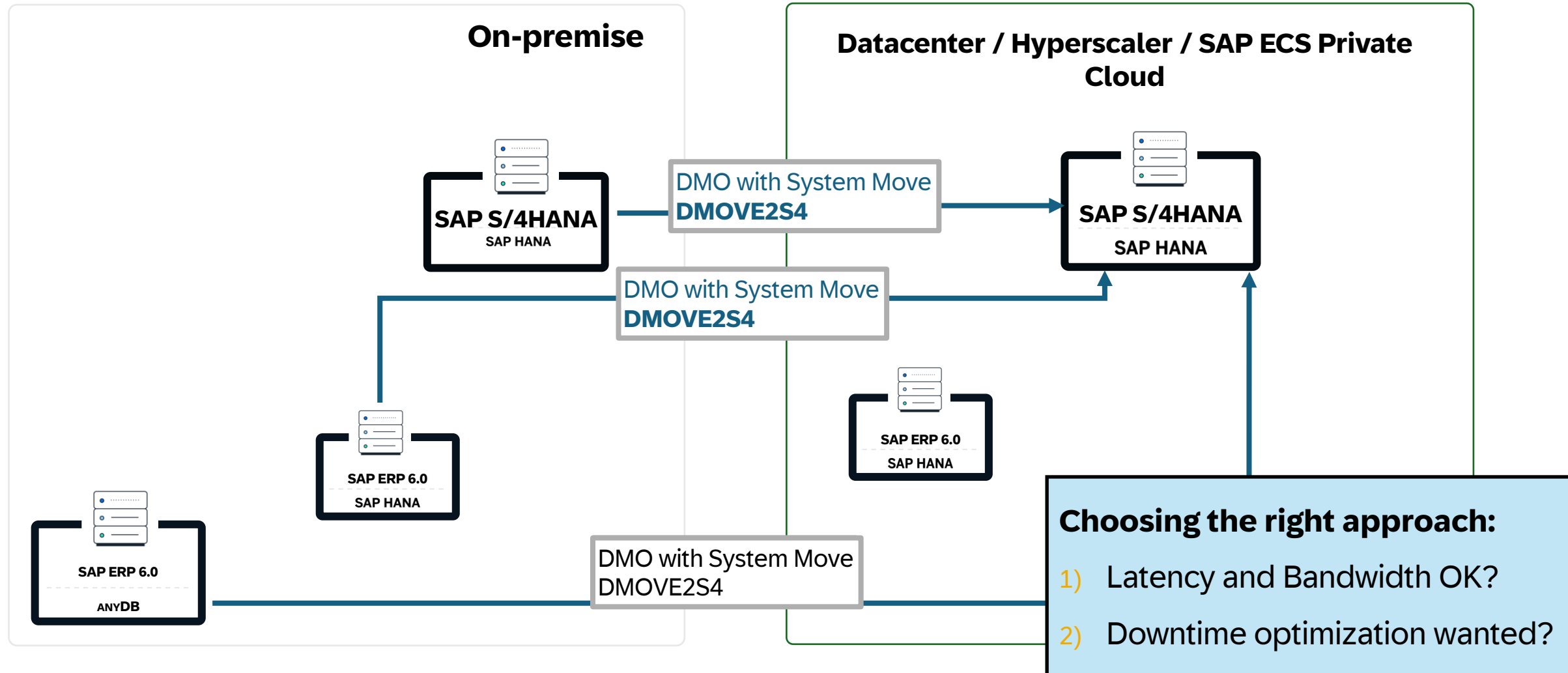
Customer Success Stories – DMOVE2S4 & doDMO

INTERNAL – SAP and Customers Only



Transition Approaches and Tools

“DMO with System Move” and “DMOVE2S4”

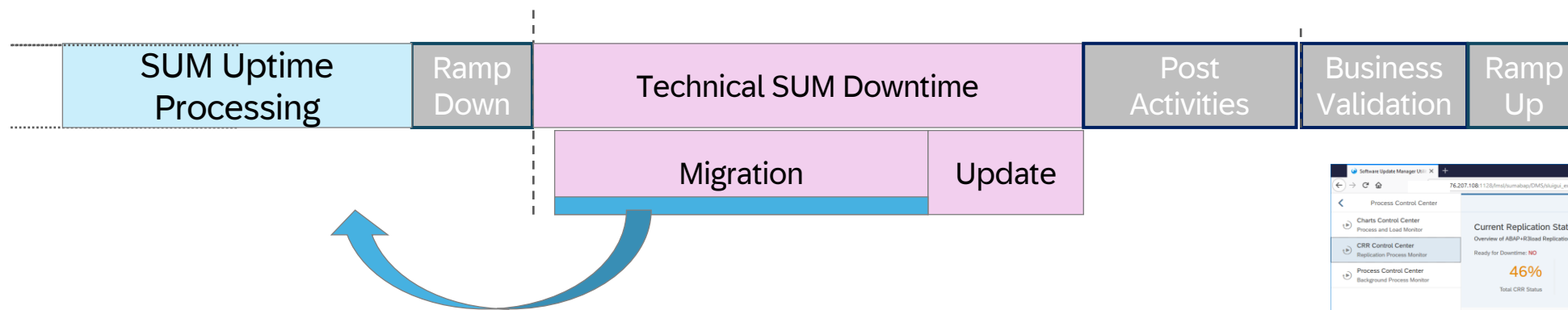


Recommendation for DMOVE2S4: latency lower than 20ms and bandwidth higher than 400 Mbps. See SAP Note [3620006](#)

Transition Approaches and Tools

Downtime-optimized Approach – **downtime-optimized DMO**

- Generally available with SUM 2.0 SP 06 (and higher) for system conversions and migrations
- **Uptime migration** for selected large application tables
 - User changes are reflected with record-and-replay technology of SUM
 - Includes initial and delta migration in uptime, remaining delta migration in downtime



- Option is offered on SUM dialog on scenario strategy
- Report available to select appropriate tables for uptime migration
- Replication monitoring is part of SUM Utilities
- Applicable only if source is not yet on SAP HANA database

The screenshot shows the 'Current Replication Status' in the SUM Utilities interface. It displays the following information:

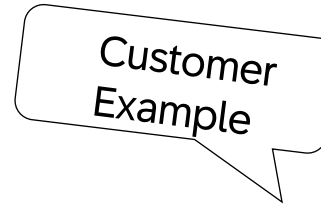
- Ready for Downtime:** NO
- Total CRR Status:** 46%
- ABAP CRR Status:** 47%
- R3load CRR Status:** 0%

Below this, there are sections for 'ABAP Replication Monitor' and 'R3load Replication Monitor', both showing 'running' status. At the bottom, there is a table with the following columns: Table Name, Replay Status, Recorded Changes, Replicated Changes, Pending Changes, and Percentage (in %).

Table Name	Replay Status	Recorded Changes	Replicated Changes	Pending Changes	Percentage (in %)
ED140		0	0	0	0
CDCLS		0	0	0	0

<https://community.sap.com/t5/technology-blogs-by-sap/dmo-downtime-optimization-by-migrating-app-tables-during-uptime/ba-p/13108040>

S/4HANA 2023 Brownfield conversion | Live Source 9 TB | 60 hours business downtime



Customer

- FMCG Industry
- Single Global S/4 HANA system

Project

- S/4 Conversion and RISE Migration
- Duration 9 months
- **Source DB Size: 8.7 TB**
- **Target DB size: 2 TB**

Technical Scope

- Source SAP Version : ECC 6.0 EHP6
- Target SAP Version : S/4 HANA 2023 FPS03
- SUM Version : SUM 2.0 SP25
- **Technical Downtime : ~11 hours using DoDMO**
- SAP landscape on RISE

Scope

- **S/4HANA Migration Scope:** Migrate ERP system to S/4HANA 2023 FPS03 with minimum changes approach
- **Fiori:** Embedded Fiori Activation

Key Figures

- **80% reduction in downtime with Downtime Optimized DMO approach**
- 3 Mocks of production copy were performed
- Migration time was brought down from 50 hours to 1.5 hours.

S/4HANA 2023 Brownfield conversion | Live
Source 18 TB | 60 hours business downtime

Customer
Example

Customer

- Utility Industry
- Single Global S/4 HANA system
- Multiple parallel Tracks

Project

- S/4 Conversion and RISE Migration
- Duration 12 months
- **Source DB Size: 18 TB(Compressed)**
- **Target DB size: 7 TB**

Technical Scope

- Source SAP Version : ECC 6.0 EHP 8
- Target SAP Version : S/4 HANA 2023 FPS03
- SUM Version : SUM 2.0 SP25
- **Technical Downtime : ~14 hours using DoDMO**
- SAP landscape on RISE

Scope

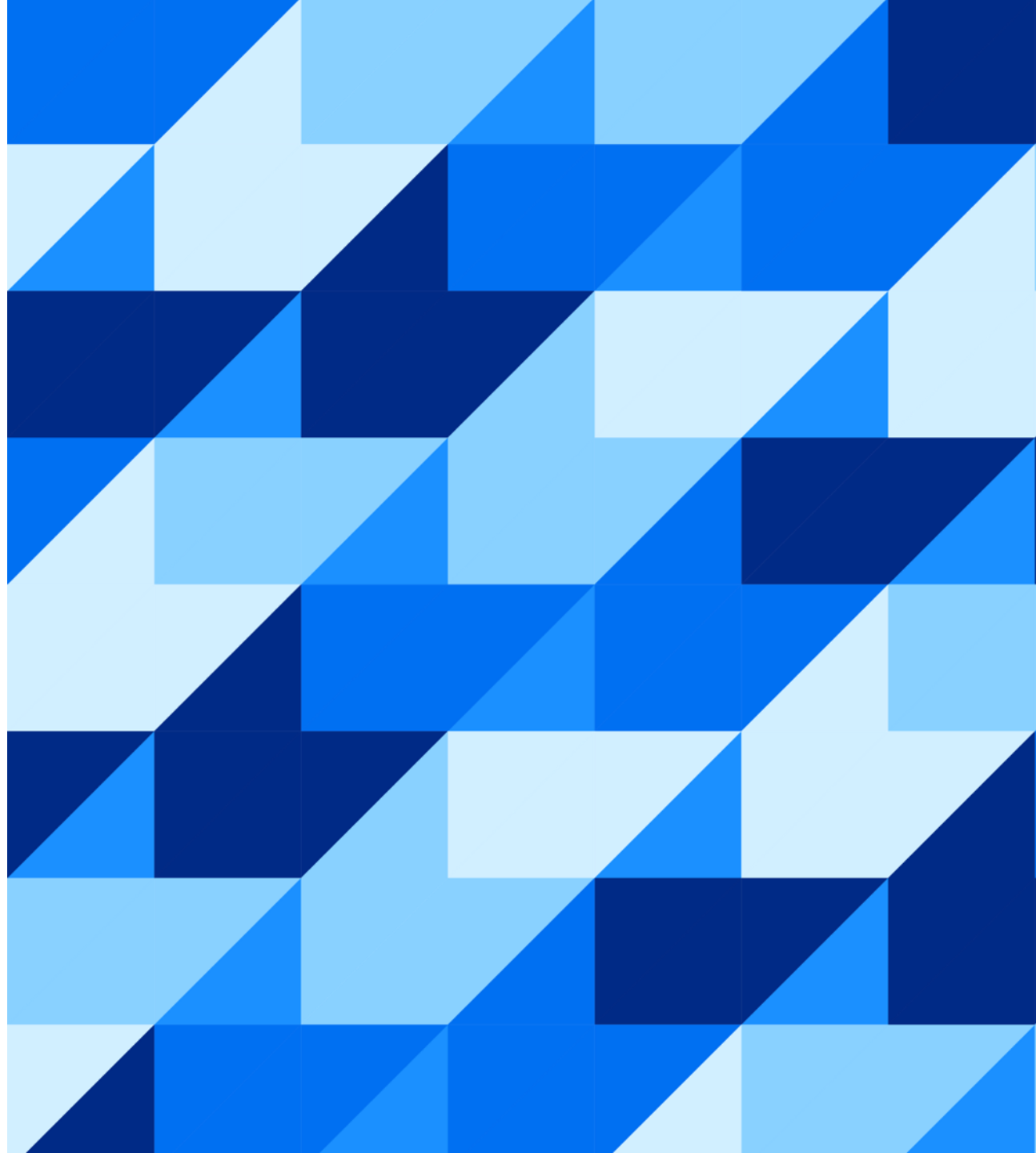
- **S/4HANA Migration Scope:** RISE Migration and S/4 HANA Conversion
- **Fiori:** Embedded Fiori Activation
- Build Work Zone Activation
- Joule Activation
- BW/4 HANA Green Field Implementation

Key Figures

- **90% reduction in downtime with Downtime Optimized DMO approach**
- 3 Mocks of production copy were performed
- Migration time was brought down from 70 hours to 4 hours.

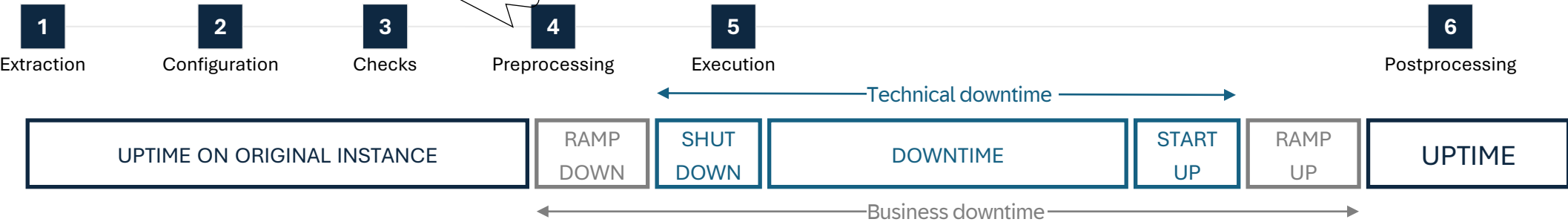
Customer Success Stories – ZDO

INTERNAL – SAP and Customers Only

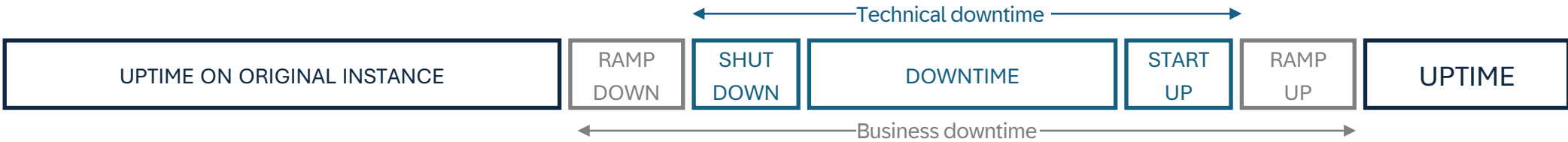


Downtimes of the different upgrade approaches

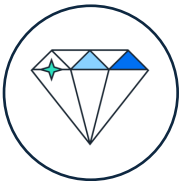
Workbench Lock



- Standard approach



- Near-Zero Downtime Maintenance (nZDM)



- Zero Downtime Option (ZDO)

Mahindra & Mahindra seamlessly performs upgrades with Zero Downtime Option of Software Update Manager

Challenge

Mahindra & Mahindra Limited operates a colossal 2 x 9TB scale-out SAP S/4HANA Private Cloud production system. To stay up-to-date, Mahindra planned to upgrade to the current major release. However, minimizing business disruption is a priority, especially in the fast-paced automotive manufacturing industry where even the slightest downtimes can have critical impact.

Solution

To minimize unavailability of the system, Mahindra leveraged the Zero Downtime Option (ZDO) of Software Update Manager. This downtime-optimization approach is offered with an additional service under the SAP Enterprise Cloud Services (ECS). Through close collaboration between SAP MaxAttention, Enterprise Cloud Services Upgrade Excellence Center and Cloud Lifecycle Management development teams, Mahindra completed a major release upgrade – from SAP S/4HANA 2020 to SAP S/4HANA 2023 – with a system unavailability of only 20 minutes.

Major achievements

- Overall business downtime of 20 hours
- Minimal disruption for Mahindra's cloud experience
- Successful ZDO handling for massive scale-out system (18TB in total)

"By leveraging ZDO, we significantly reduced the downtime compared to our previous upgrades, which typically required a 48-hour window. The project's success can be attributed to the exceptional collaboration among the Mahindra & Mahindra and SAP teams."

Sanjay Chavan, VP & Head of Enterprise Applications, Mahindra & Mahindra

• **20 mins**

- Unavailability of the system for a major release upgrade

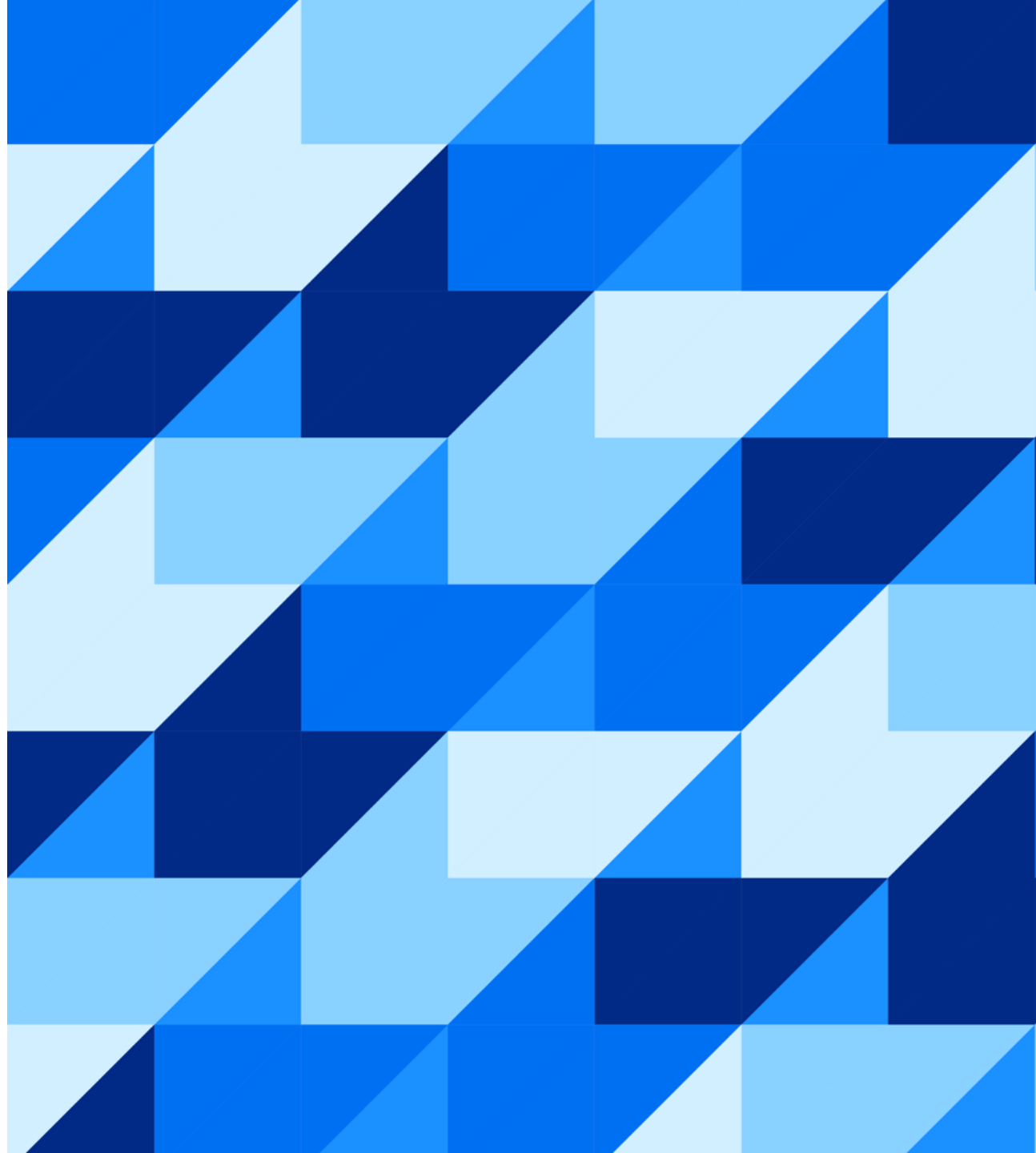
• **18 TB**

- Size of the S/4HANA production system that was successfully upgraded using ZDO

• Company Name	• Industry	• Product and Services	• Employees	• Revenue	• Featured solutions
• Mahindra & Mahindra Limited https://www.mahindra.com/	• Automotive	• Passenger vehicles • Military vehicles • Digital Farming Technologies	• >140,500	• US\$16 billion	• Zero Downtime Option of Software Update Manager

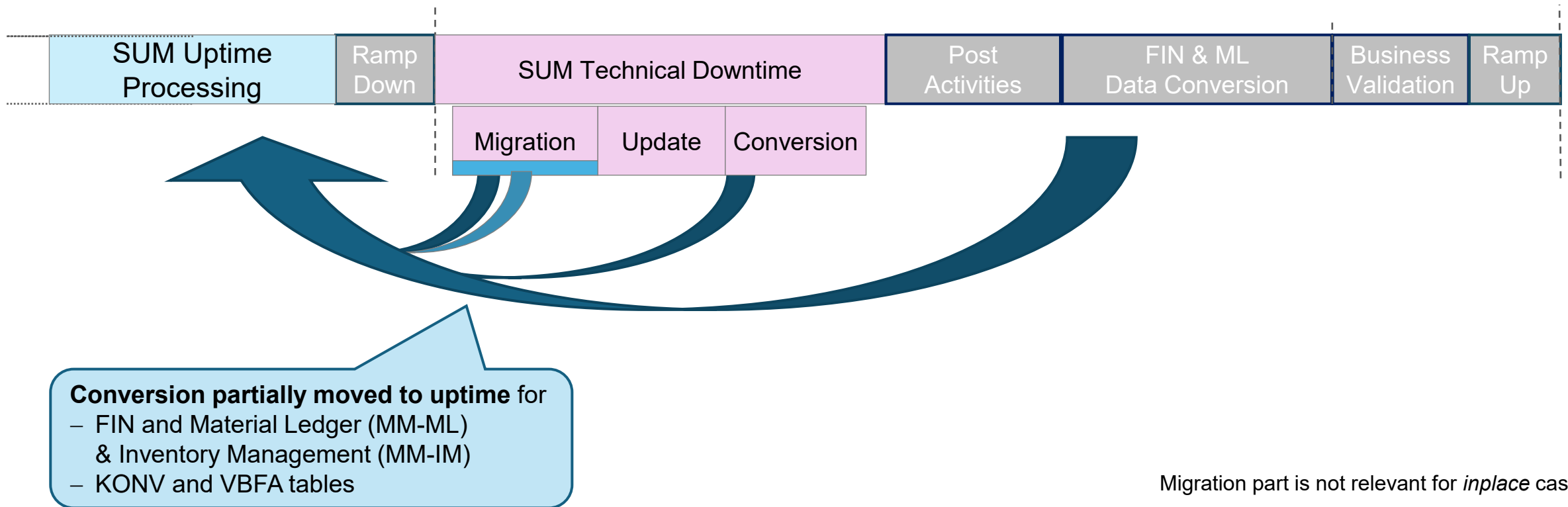
Customer Success Stories – doC (CustOnTemp)

INTERNAL – SAP and Customers Only



Downtime-optimized conversion approach

- **Data conversion** moved to uptime processing, also for FIN & ML
- **Field conversion** moved to uptime processing (KONV and VBFA tables)
- **Uptime migration** for selected large application tables (which are not part of the data conversion)

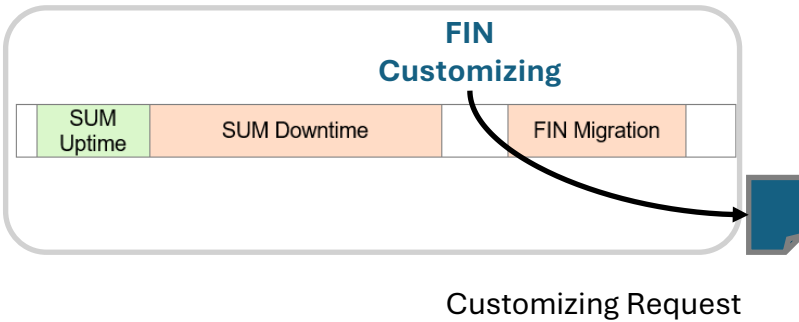


Migration part is not relevant for *inplace* case

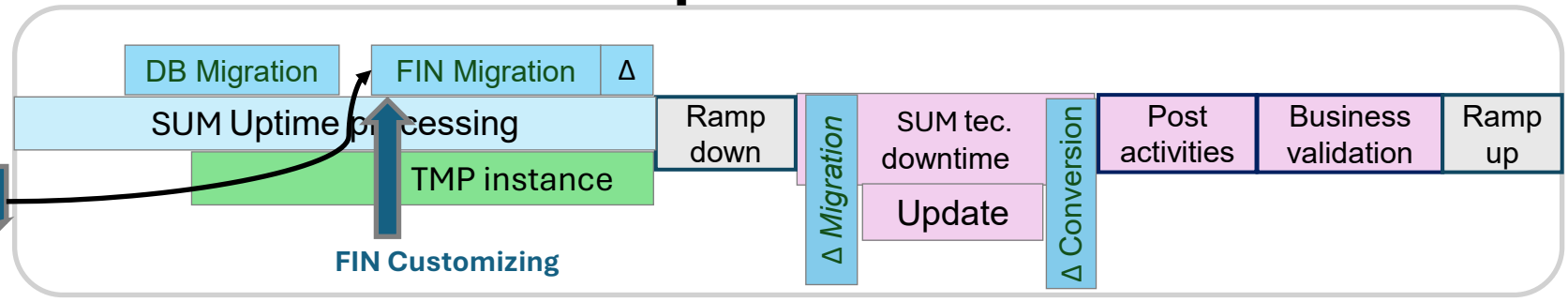
Downtime-Optimized Conversion with CustOnTmp

“Customizing on Temporary Instance” in a Nutshell

Standard Conversion



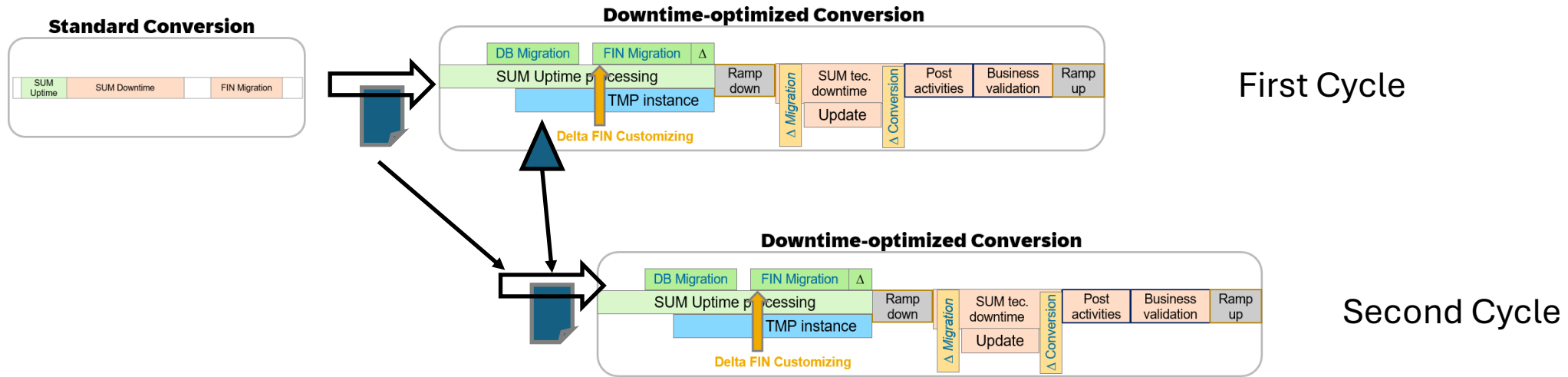
Downtime-optimized Conversion



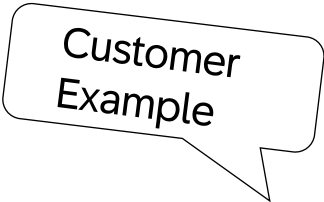
- The downtime-optimized Conversion approach executes **FIN data migration** already in uptime
- This requires the **FIN customizing** (on S/4HANA-level) to be provided beforehand, created by a standard conversion run prior to the downtime-optimized conversion run
- This requirement of „always a standard conversion run beforehand“ is eliminated by *CustOnTmp*: the FIN customizing can be adapted **during** the downtime-optimized conversion run
- SUM always creates a second instance (here: temporary instance, TMP) on target level
- This TMP instance is extended to enable customizing on target level, parallel to original instance

Downtime-Optimized Conversion with CustOnTmp

Simplified Procedure



- One standard conversion run will be done to create the initial FIN customizing
- All subsequent downtime-optimized conversion runs consume this FIN customizing
- Any FIN customizing changes in PRD after the standard run are reflected with delta customizing activity on the TMP instance



Customer

- Entertainment Industry
- Single Global S/4 HANA system

Project

- S/4 Conversion
- Duration 29 months
- Source DB size: 38TB
- Target DB Size : 11.7TB

Technical Scope

- Source SAP Version : ECC 6.0 EHP7
- Target SAP Version : S/4 HANA 2022 FPS03
- SUM Version : SUM 2.0 SP19
- Technical Downtime : 15 hours using DoC
- SAP landscape on premise

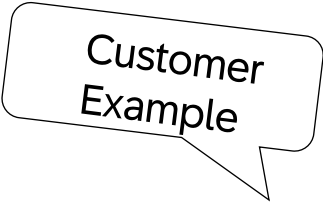
Scope

- **S/4HANA Migration Scope:** Convert ERP system to S/4HANA 2022 with minimum changes approach
- **Fiori:** Embedded Fiori Activation

Key Figures

- **85% reduction in the downtime with DoC-CustOnTmp approach**
- 7 Mocks of production copy were performed
- 2.6 Billion delta records replicated in uptime.

S/4HANA 2023 Brownfield conversion | Live
Source 17 TB | 60 hours business downtime



Customer

- FMCG Industry
- Single Global S/4 HANA system

Project

- S/4 Conversion
- Duration 18 months
- Source DB Size: 17TB
- Target DB size: 4.5TB

Technical Scope

- Source SAP Version : ECC 6.0 EHP7
- Target SAP Version : S/4 HANA 2023 FPS01
- SUM Version : SUM 2.0 SP22
- Technical Downtime : ~8 hours using DoC
- SAP landscape on premise

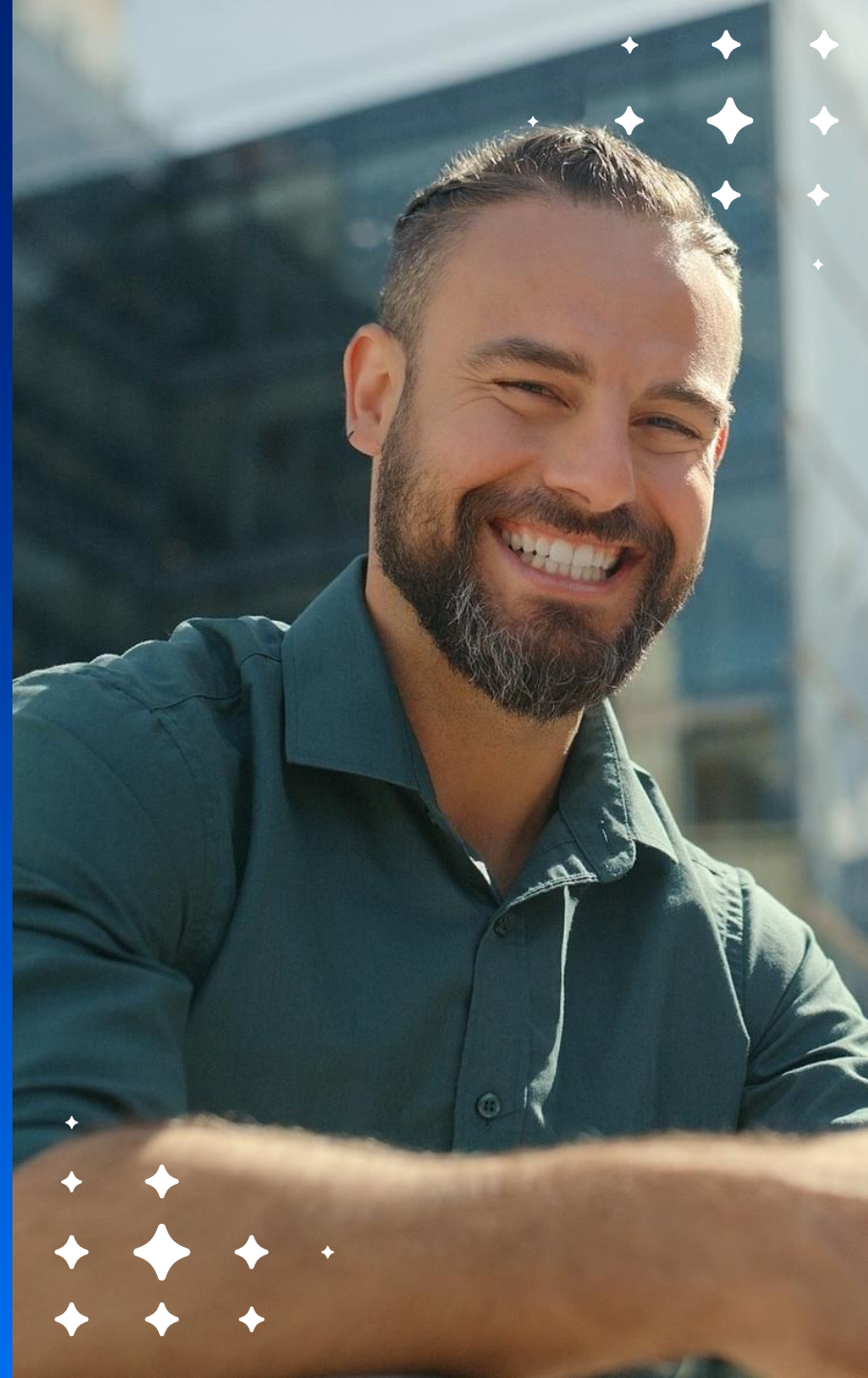
Scope

- **S/4HANA Migration Scope:** Migrate ERP system to S/4HANA 2023 with minimum changes approach
- **Fiori:** Embedded Fiori Activation

Key Figures

- **85% reduction in downtime with DoC – CustOnTmp approach**
- 7 Mocks of production copy were performed
- ~500 million delta records replicated in uptime

AI in Action



Ambition migration and modernization



Approximately

35%

reduction of migration effort with an Agent-led toolchain – and a path to mostly autonomous transitions

Step 1

2026:

AI-supported migration

Providing AI capabilities and agents that reduce customer effort significantly when moving to Private Cloud via RISE, packaged and commercialized as one platform

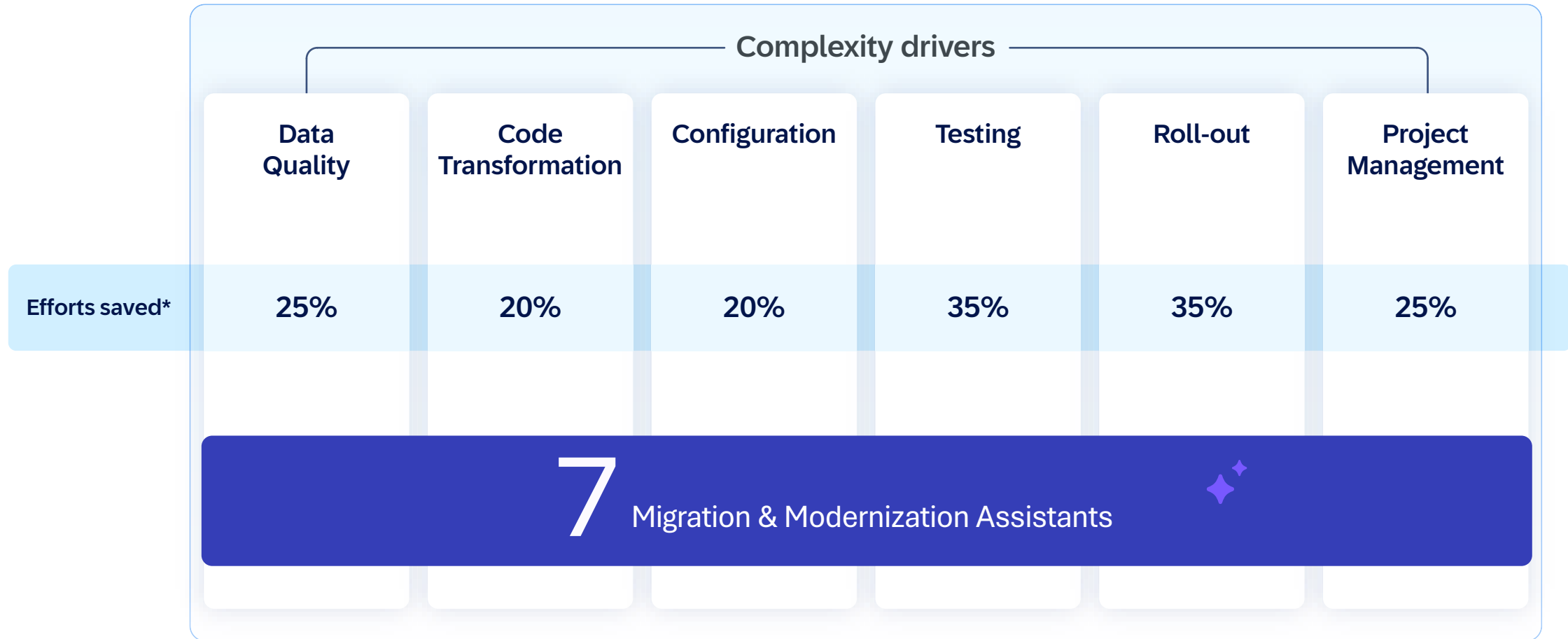
Step 2

Long-shot:

Nearly automated migration

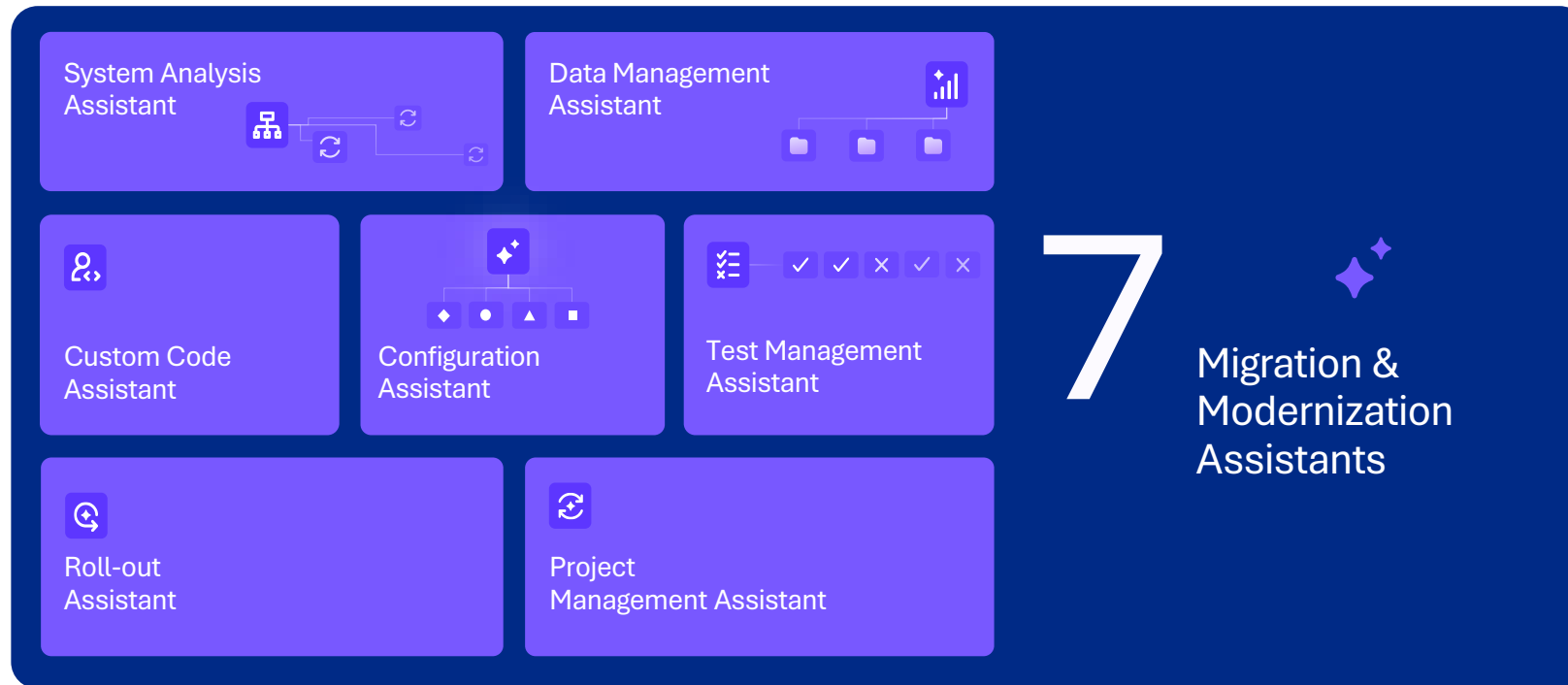
Build an autonomous migration fabric: Intelligent agents that continuously analyze legacy estates – data models, customizations, and extensions – then automatically design, configure, test, and execute end-to-end migrations to the cloud solutions with minimal human touch

Migration effort is driven by six complexity drivers

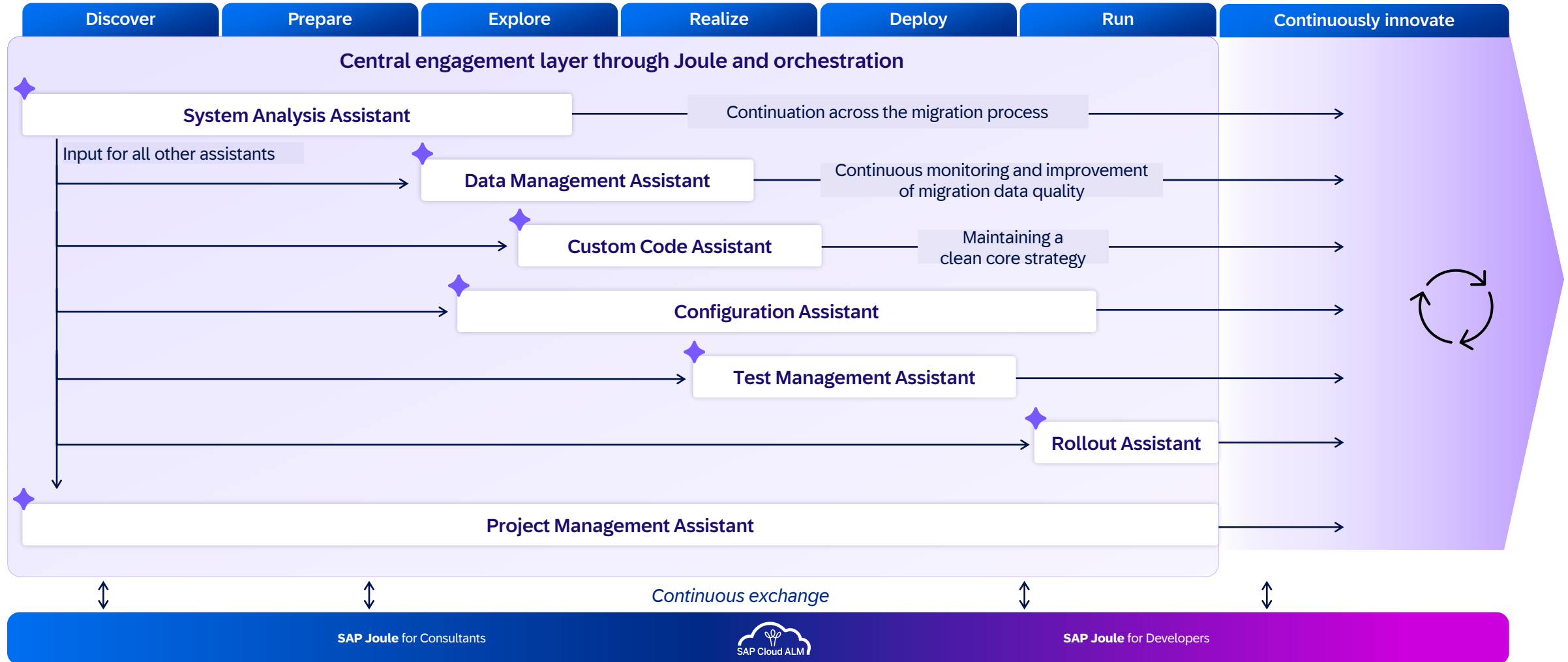


*Effort reduction within the activities addressed by AI cases (e.g. 35% effort savings in test automation refers to all testing activities)

Introducing seven migration and modernization assistants



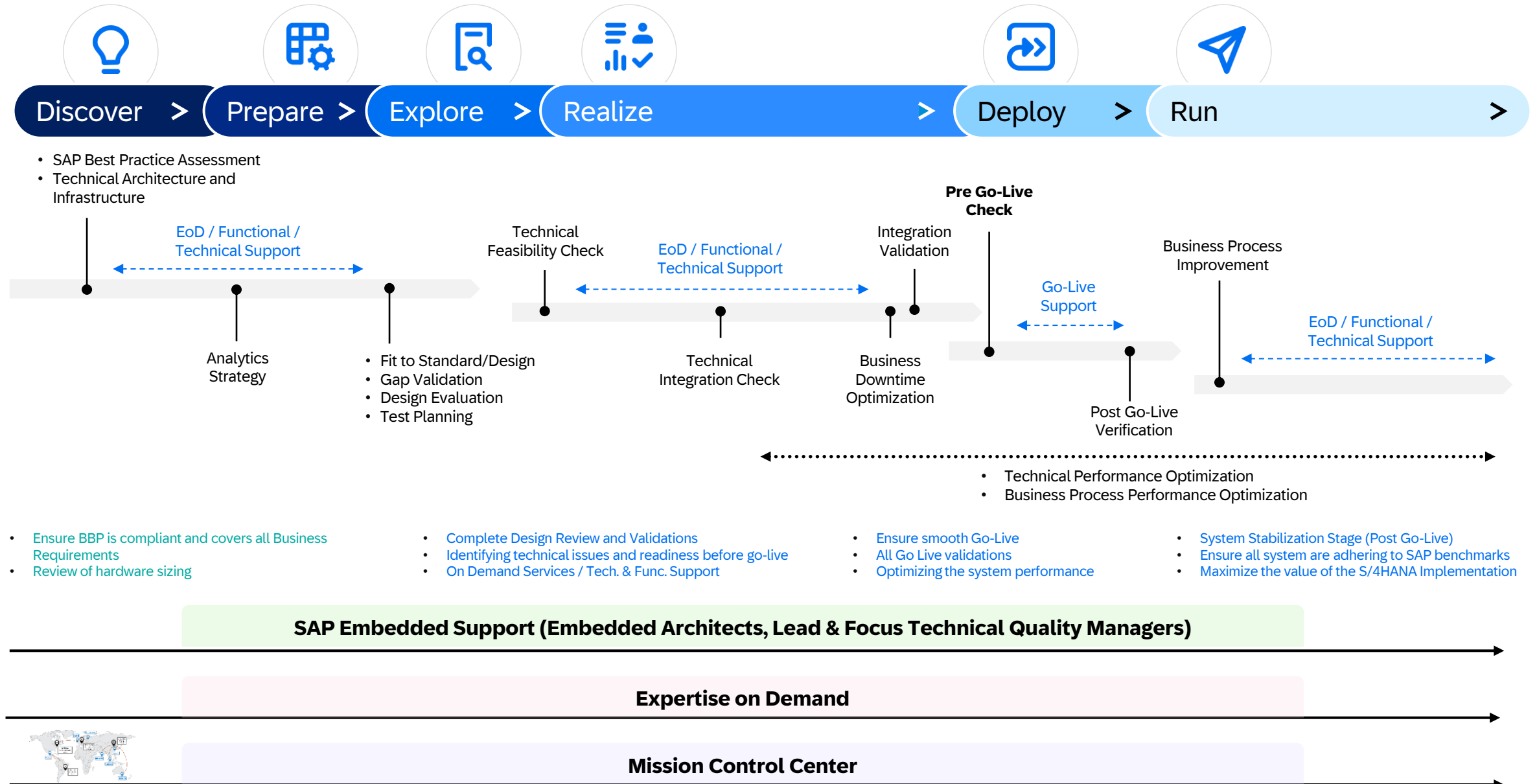
Assistants supporting ongoing modernization ✨



SAP Services



SAP Services for S/4HANA conversions & RISE migration



Contact Information:

nikhil.Shetty@sap.com

abhishek.srivastava@sap.com

abhishek.bhatt@sap.com